

Problem Set 03
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```
init: starting sh
$ $ pgtbltest
print_ptbl starting
va 0x0 pte 0x21FC7C5B pa 0x87F1F000 perm 0x5B
va 0x1000 pte 0x21FC701B pa 0x87F1C000 perm 0x1B
va 0x2000 pte 0x21FC6C17 pa 0x87F1B000 perm 0x17
va 0x3000 pte 0x21FC6807 pa 0x87F1A000 perm 0x7
va 0x4000 pte 0x21FC64D7 pa 0x87F19000 perm 0xD7
va 0x5000 pte 0x0 pa 0x0 perm 0x0
va 0x6000 pte 0x0 pa 0x0 perm 0x0
va 0x7000 pte 0x0 pa 0x0 perm 0x0
va 0x8000 pte 0x0 pa 0x0 perm 0x0
va 0x9000 pte 0x0 pa 0x0 perm 0x0
va 0xFFFF6000 pte 0x0 pa 0x0 perm 0x0
va 0xFFFF7000 pte 0x0 pa 0x0 perm 0x0
va 0xFFFF8000 pte 0x0 pa 0x0 perm 0x0
va 0xFFFF9000 pte 0x0 pa 0x0 perm 0x0
va 0xFFFFA000 pte 0x0 pa 0x0 perm 0x0
va 0xFFFFB000 pte 0x0 pa 0x0 perm 0x0
va 0xFFFFC000 pte 0x0 pa 0x0 perm 0x0
va 0xFFFFD000 pte 0x21FD4013 pa 0x87F50000 perm 0x13
va 0xFFFFE000 pte 0x21FD48C7 pa 0x87F52000 perm 0xC7
va 0xFFFFF000 pte 0x2000184B pa 0x80006000 perm 0x4B
print_ptbl: OK
ugetpid_test starting
ugetpid_test: OK
print_kpgtbl starting
page table 0x0000000087f22000
..0x0000000000000000: pte 0x0000000021fc7801 pa 0x0000000087f1e000
.. ..0x0000000000000000: pte 0x0000000021fc7401 pa 0x0000000087f1d000
.. .. ..0x0000000000000000: pte 0x0000000021fc7c5b pa 0x0000000087f1f000
.. .. ..0x00000000000001000: pte 0x0000000021fc701b pa 0x0000000087f1c000
.. .. ..0x00000000000002000: pte 0x0000000021fc6cd7 pa 0x0000000087f1b000
.. .. ..0x00000000000003000: pte 0x0000000021fc6807 pa 0x0000000087f1a000
.. .. ..0x00000000000004000: pte 0x0000000021fc64d7 pa 0x0000000087f19000
..0x00000003fc0000000: pte 0x0000000021fc8401 pa 0x0000000087f21000
.. ..0x00000003ffffe00000: pte 0x0000000021fc8001 pa 0x0000000087f20000
.. .. ..0x00000003fffffd000: pte 0x0000000021fd4013 pa 0x0000000087f50000
.. .. ..0x00000003fffffe000: pte 0x0000000021fd48c7 pa 0x0000000087f52000
.. .. ..0x00000003ffffff000: pte 0x000000002000184b pa 0x0000000080006000
print_kpgtbl: OK
superpg_test starting
superpg_test: OK
pgtbltest: all tests succeeded
$
```

```
usertrap(): unexpected scause 0xf pid=6538
      sepc=0x1c8c stval=0x1000000000000000
usertrap(): unexpected scause 0xf pid=6539
      sepc=0x1c8c stval=0x2000000000000000
usertrap(): unexpected scause 0xf pid=6540
      sepc=0x1c8c stval=0x4000000000000000
usertrap(): unexpected scause 0xf pid=6541
      sepc=0x1c8c stval=0x8000000000000000
OK
test sbrkfail: usertrap(): unexpected scause 0xd pid=6553
      sepc=0x3a52 stval=0x13000
OK
test sbrkarg: OK
test validatetest: OK
test bsstest: OK
test bigargtest: OK
test argptest: OK
test stacktest: usertrap(): unexpected scause 0xd pid=6561
      sepc=0x1ce0 stval=0x10e90
OK
test nowrite: usertrap(): unexpected scause 0xf pid=6563
      sepc=0x1d86 stval=0x0
usertrap(): unexpected scause 0xf pid=6564
      sepc=0x1d86 stval=0x80000000
usertrap(): unexpected scause 0xf pid=6565
      sepc=0x1d86 stval=0x3fffffe000
usertrap(): unexpected scause 0xf pid=6566
      sepc=0x1d86 stval=0x3ffffff000
usertrap(): unexpected scause 0xf pid=6567
      sepc=0x1d86 stval=0x4000000000
usertrap(): unexpected scause 0xf pid=6568
      sepc=0x1d86 stval=0xffffffffffffff
OK
test pgbug: OK
test sbrkbugs: usertrap(): unexpected scause 0xc pid=6571
      sepc=0x4b24 stval=0x4b24
usertrap(): unexpected scause 0xc pid=6572
      sepc=0x4b24 stval=0x4b24
OK
test sbrklast: OK
test sbrk8000: OK
test badarg: OK
ALL TESTS PASSED
$ QEMU: Terminated
```

```

no-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie -c -o kernel/spinlock.o kernel/spinlock.c
riscv64-linux-gnu-gcc -Wall -Werror -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -DSQL_PGTBL -DLAB_PGTBL -MD -mmodel=medany -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-str
t -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-bui
no-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie -march=rv64g -nostdinc -I. -Ikernl -c user/initcode.S -o user/initcode.o
riscv64-linux-gnu-ld -z max-page-size=4096 -N -e start -Ttext 0 -o user/initcode.out user/initcode.o
riscv64-linux-gnu-ld: warning: user/initcode.out has a LOAD segment with RWX permissions
riscv64-linux-gnu-objcopy -S -O binary user/initcode.out user/initcode
riscv64-linux-gnu-objdump -S user/initcode.o > user/initcode.asm
riscv64-linux-gnu-ld -z max-page-size=4096 -T kernel/kernel.ld -o kernel/kernel kernel/entry.o kernel/kalloc.o kernel/string.o kernel/main.o kernel/vm.o kernel/proc.o kernel/switch.o
scall.o kernel/sysproc.o kernel/bio.o kernel/fs.o kernel/log.o kernel/sleeplock.o kernel/file.o kernel/pipe.o kernel/exec.o kernel/sysfile.o kernel/kernelvec.o kernel/plc.o kernel/
o kernel/print.o kernel/uart.o kernel/spinlock.o
riscv64-linux-gnu-ld: warning: kernel/kernel has a LOAD segment with RWX permissions
riscv64-linux-gnu-objdump -S kernel/kernel > kernel/kernel.asm
riscv64-linux-gnu-objdump -t kernel/kernel | sed '1,/SYMBOL TABLE/d; s/ .* / /; /$/d' > kernel/kernel.sym
make[1]: Leaving directory '/home/tayyaba/xv6-labs-2026'
== Test pgtbltest ==
$ make qemu-gdb
(16.1s)
== Test pgtbltest: ugetpid ==
pgtbltest: ugetpid: OK
== Test pgtbltest: print_kpgtbl ==
pgtbltest: print_kpgtbl: OK
== Test pgtbltest: superpg ==
pgtbltest: superpg: OK
== Test answers-pgtbl.txt ==
answers-pgtbl.txt: OK
== Test usertests ==
$ make qemu-gdb
Timeout! (301.3s)
== Test usertests: all tests ==
usertests: all tests: FAIL
...
test exitiput: OK
test iput: OK
test opentest: OK
test writetest: OK
test writebig: qemu-system-riscv64: terminating on signal 15 from pid 18474 (make)
MISSING: 'ALL TESTS PASSED$'
== Test time ==
time: OK
Score: 41/51
make: *** [Makefile:345: grade] Error 1
tayyaba@tayyaba-VirtualBox: /xv6-labs-2026$ git status

```

usertests timed out at 300s because the VirtualBox VM is slow, but **usertests -q** passes manually (ALL TESTS PASSED).